

QUESTION BANK

CHAPTER: ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

Section A

1 Mark Questions

Q1. A coil of N turns and area A is rotated at the rate of n rotations per second in a magnetic field of intensity B , the magnitude of the maximum magnetic flux will be

- (a) NAB (b) nAB (c) $NnAB$ (d) $2\pi nNAB$

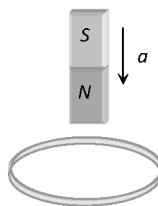
Q2. In electromagnetic induction, the induced e.m.f. in a coil is independent of

- (a) Change in the flux (b) Time (c) Resistance of the circuit (d) none of the above

Q3. Lenz's law is consequence of the law of conservation of

- (a) Charge (b) Momentum (c) Mass (d) Energy

Q4. A metallic ring is attached with the wall of a room. When the north pole of a magnet is brought near to it, the induced current in the ring will be



- (a) First clockwise then anticlockwise (b) In clockwise direction
(c) In anticlockwise direction (d) First anticlockwise then clockwise

Q5. Lenz's law is expressed by the following formula (here e = induced e.m.f., ϕ = magnetic flux in one turn and N = number of turns)

- (a) $e = -\phi \frac{dN}{dt}$ (b) $e = -N \frac{d\phi}{dt}$ (c) $e = -\frac{d}{dt} \left(\frac{\phi}{N} \right)$ (d) $e = N \frac{d\phi}{dt}$

Q6. A moving conductor coil in a magnetic field produces an induced e.m.f. This is in accordance with

- (a) Amperes law (b) Coulomb law (c) Lenz's law (d) Faraday's law

Q7. The self-inductance of a coil is 5 henry, a current of 1 amp change to 2 amp within 5 second through the coil. The value of induced e.m.f. will be

- (a) 10 volt (b) 0.10 volt (c) 1.0 volt (d) 100 volt

Q8. The self-inductance of a solenoid of length L , area of cross-section A and having N turns is

- (a) $\frac{\mu_0 N^2 A}{L}$ (b) $\frac{\mu_0 NA}{L}$ (c) $\mu_0 N^2 LA$ (d) $\mu_0 NAL$

Q9. A magnet is brought towards a coil (i) speedily (ii) slowly, then the induced e.m.f will be respectively

- (a) More in first case and more in first case
 (b) More in first case and equal in both case
 (c) Less in first case and more in second case
 (d) Less in first case and equal in both case

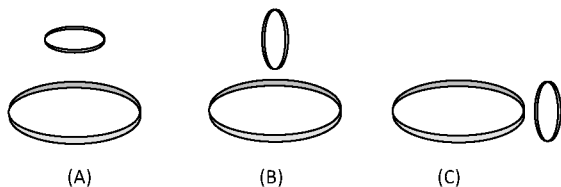
Q10. The direction of induced e.m.f. during electromagnetic induction is given by

- (a) Faraday's law (b) Lenz's law (c) Maxwell's law (d) Ampere's law

Q11. The unit of magnetic flux is

- (a) weber/m² (b) Weber (c) Henry (d) Ampere/m

Q12. Two circular coils can be arranged in any of the three situations shown in the figure. Their mutual inductance will be



- (a) Maximum in situation (A) (b) Maximum in situation (B)
 (c) Maximum in situation (C) (d) The same in all situations

Assertion Reason Questions

Directions: (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

- (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.

Q13. **Assertion** : Magnetic flux can produce induced e.m.f.

Reason: Lenz established induced e.m.f. experimentally.

Q14. **Assertion** : An induced emf is generated when magnet is withdrawn from the solenoid.

Reason: The relative motion between magnet and solenoid induces emf.

Q15. **Assertion** : A transformer cannot work on dc supply.

Reason: dc changes neither in magnitude nor in direction.

Q16. **Assertion** : . EMF induce in coil when flux link with coil changes

Reason: It induce to oppose the cause which produce It

Q17. **Assertion** : Magnetic flux link with coil is directly proportional to current.

Reason: Self-inductance will not play any role in linking of magnetic flux with coil.

Q18. **Assertion** : Lenz law is based on conservation of energy.

Reason : Constant current is able to induce emf in coil.

Q19. What is the SI unit of magnetic flux?

Q20. What is Faraday law electromagnetic induction?

2-Marks Questions

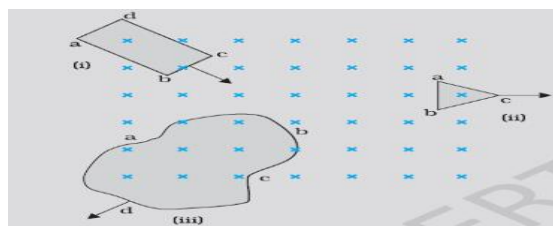
Q21. State and Explain Lenz's law?

Q22. What are the factors on which self-inductance depends?

3-Marks Questions

Q23. Define the coefficient of self-inductance (L). Write its unit. Write the two factors on which the self-inductance of along solenoid depends.

Q24. Figure shows planar loops of different shapes moving out of or into a region of a magnetic field which is directed normal to the plane of the loop away from the reader. Determine the direction of induced current in each loop using Lenz's law.



Q25. Obtain an expression for the self-inductance of a long solenoid. Hence define one henry.

Section A

Answer Key

1. (a) Since $\phi = NBA \cos\theta$; For ϕ to be maximum; $\cos\theta = \max = 1$ so $\phi_{\max} = NBA$.
2. (c) Because induced e.m.f. is given by $E = -N \frac{d\phi}{dt}$.
3. (d) The energy of the field increases with the magnitude of the field. Lenz's law infers that there is an opposite field created due to increase or decrease of magnetic flux around a conductor so as to hold the law of conservation of energy.
4. (c) As it is seen from the magnet side induced current will be anticlockwise.
5. (b)
6. (d)
7. (c) $e = -L \frac{di}{dt} \Rightarrow e = 5 \times \frac{1}{5} = 1 \text{ volt}$
8. (a)
9. The magnitude of induced e.m.f. is directly proportional to the rate of change of magnetic flux. Induced charge doesn't depend upon time.
10. (b)
11. (b)
12. (a) As maximum linkage of magnetic flux is possible in A situation .
13. (d) E.M.F. induces, when there is change in magnetic flux. Faraday did experiment in which, there is relative motion between the coil and magnet, the flux linked with the coil changes and e.m.f. induces.
14. (a) as there is relative motion between solenoid and magnet gives change in flux
15. (a) Transformer works on ac only, ac changes in magnitude as well as in direction.
16. (a)
17. (c)
18. (c)
19. The SI unit of magnetic flux is weber or tesla – metre
20. Whenever the magnetic flux linked with an electric circuit changes an e.m.f. is induced in the circuit. The phenomenon is called electromagnetic induction.
21. The direction of induced current due to the change in magnetic flux through a closed-loop is always such that it opposes the change or cause which produces it.

when Lenz law is applied to it, it will be

$E \propto -d\Phi/dt$ where a negative sign shows the opposing nature of induced emf. It is based on the principle of conservation of energy.

22. $L = (\mu_0 N^2 A) / l$

- It depends on number of turns N
- It depends on area of cross section A
- Permeability of core material μ_0

23. From self-induction, we know that if I is the strength of the current flowing through a coil at any time ϕ is amount of magnetic flux linked with the coil at that time. It is found that

$$\phi \propto I, \phi \propto LI \quad \dots(i)$$

L is a constant of proportionality and is called coefficient of self induction or self inductance of the coil.

S.I Unit $\frac{\text{volt sec}}{\text{Amp}}$ or Henry

The value of L depends upon the number of turn of the coil, area of cross-section and nature of material of the core on which coil is wound.

24. (i) The magnetic flux through the rectangular loop abcd increases, due to the motion of the loop into the region of magnetic field, The induced current must flow along the path bcdab so that it opposes the increasing flux.
- (ii) Due to the outward motion, magnetic flux through the triangular loop abc decreases due to which the induced current flows along bacb, so as to oppose the change in flux.
- (iii) As the magnetic flux decreases due to motion of the irregular shaped loop abcd out of the region of magnetic field, the induced current flows along cdabc, so as to oppose change in flux.

Note that there are no induced current as long as the loops are completely inside or outside the region of the magnetic field.

25. Consider current I flowing through a long solenoid of area A,

Let N be the total number of turns in the solenoid,

Total flux, $\phi = NBA$

Here, $B = \mu_0 n I$

Where, n is no. of turns per unit length of the solenoid, $N = n l$

$$\phi = n l \times \mu_0 n I A$$

$$\Rightarrow \phi = \mu_0 n^2 A l I \quad \dots\dots (1)$$

$$\text{Also, } \phi = L I \quad \dots\dots (2)$$

From equation (1) & (2)

$$I \mu_0 n^2 A l = L I$$

$$L = \mu_0 n^2 A l$$

$$L = \mu_0 N^2 A \quad \text{where } (n = N/l)$$

One henry (1H) can be defined as: If current is changing at a rate of 1A/s in a coil inducing an emf of 1 volt in it, then the inductance of the coil is one henry.

SECTION B

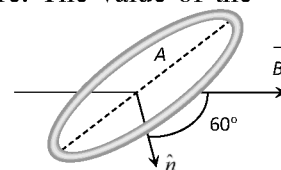
1-Mark Questions

Q1. A coil of area 100 cm^2 has 500 turns and magnetic field of 0.1 Wb/m^2 is perpendicular to the coil. The field is reduced to zero in 0.1 second . The induced e.m.f. in the coil is

- (a) 1 V (b) 5 V (c) 50 V (d) Zero

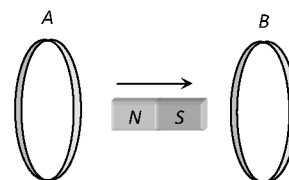
Q2. A coil of area $A = 2 \text{ m}^2$ is situated in a uniform magnetic field $B = \sqrt{3}.0 \text{ wb/m}^2$ and area vector makes an angle of 30° with respect to the magnetic field as shown in figure. The value of the magnetic flux through the area A would be equal to

- (a) 2 weber (b) 1 weber (c) 3 weber (d) $\frac{3}{2} \text{ weber}$



Q3. In the diagram shown if a bar magnet is moved along the common axis of two single turn coils A and B in the direction of arrow

- (a) Current is induced only in A & not in B
(b) Induced currents in A & B are in the same direction
(c) Current is induced only in B and not in A
(d) Induced currents in A & B are in opposite directions



Q4. Magnetic flux ϕ (in weber) linked with a closed circuit of resistance 10 ohm varies with time t (in seconds) as

$$\phi = 5t^2 - 4t + 1$$

The induced electromotive force in the circuit at $t = 0.2 \text{ sec}$. is

- (a) 0.4 volts (b) -0.4 volts (c) -2.0 volts (d) 2.0 volts

ASSERTION AND REASON

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.

Q5. Assertion : The induced e.m.f. will be same and current will be different in two identical loops of copper and aluminium, when rotated with same speed in the same magnetic field.

Reason : Induced e.m.f. is proportional to rate of change of magnetic field while induced current depends on resistance of wire.

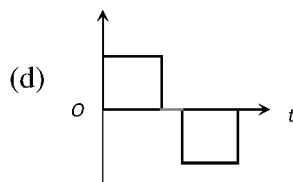
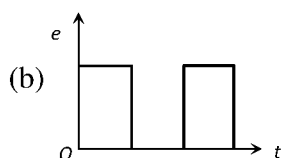
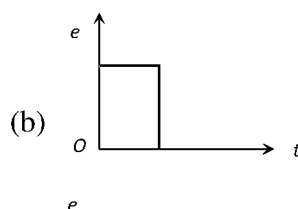
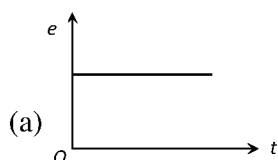
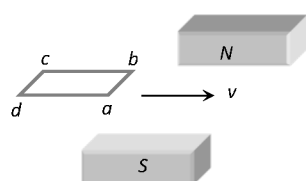
Q6. Assertion: In the phenomenon of mutual induction, self-induction of each of the coils persists.

Reason: Self-induction arises when strength of current in same coil changes. In mutual induction, current is changing in both the individual coils.

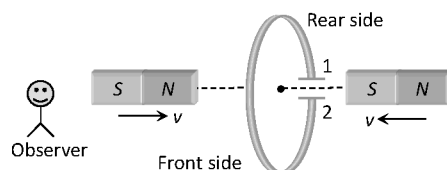
Q7. Assertion : Only a change in magnetic flux will maintain an induced current the coil.

Reason : The presence of large magnetic flux through a coil maintains a current in the coil.

Q8. A horizontal loop $abcd$ is moved across the pole pieces of a magnet as shown in fig. with a constant speed v . When the edge ab of the loop enters the pole pieces at time $t = 0 \text{ sec}$. Which one of the following graphs represents correctly the induced emf in the coil



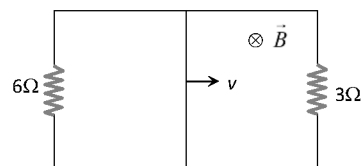
Q9. The north and south poles of two identical magnets approach a coil, containing a condenser, with equal speeds from opposite sides. Then



- (a) Plate 1 will be negative and plate 2 positive
- (b) Plate 1 will be positive and plate 2 negative
- (c) Both the plates will be positive
- (d) Both the plates will be negative

Q10. A rectangular loop with a sliding connector of length $l = 1.0 \text{ m}$ is situated in a uniform magnetic field $B = 2 \text{ T}$ perpendicular to the plane of loop. Resistance of connector is $r = 2 \Omega$. Two resistance of 6Ω and 3Ω are connected as shown in figure. The external force required to keep the connector moving with a constant velocity $v = 2 \text{ m/s}$ is

- (a) 6 N
- (b) 4 N
- (c) 2 N
- (d) 1 N



Q11. What is the principle used in the A.C. generator?

2-Marks Questions

Q12. Two identical circular loops of metal wire are lying on a table without touching each other. Loop-A carries a current which increases with time. Then what type of force is present between them?

Q13. If a square loop of conducting material is moved with a constant velocity fully inside a uniform magnetic field perpendicular to the field, will the current be induced in it?

3-Marks Questions

Q14. Current in a circuit falls from 5.0 A to 0.0 A in 0.1 s . If an average emf of 200 V induced, give an estimate of the self-inductance of the circuit.

Q15. What is mutual induction. Derive an expression for mutual induction between two coils.

Section B (Answer Key)

$$1. (b) \quad e = -\frac{N(B_2 - B_1)A \cos \theta}{\Delta t}$$

$$= -\frac{500 \times (0 - 0.1) \times 100 \times 10^{-4} \cos 0}{0.1} = 5 \text{ V}$$

2.(b) Angle between normal to the plane of the coil and direction of magnetic field is θ

$$\therefore \text{Flux linked with coil } \phi = BA \cos \theta = \sqrt{3} \cdot 0 \times 2 \times \cos 30^\circ \Rightarrow \phi = 3 \text{ weber}$$

3. (d)

$$4. (d) \quad e = -\frac{d\phi}{dt} = -(10t - 4) \Rightarrow (e)_{t=2} = -(10 \times 0.2 - 4) = 2 \text{ volt}$$

5. (a) Since both the loops are identical (same area and number of turns) and moving with a same speed in same magnetic field. Therefore same emf is induced in both the coils. But the induced current will be more in the copper loop as its resistance will be lesser as compared to that of the aluminium loop.

6. (b) Mutual inductance is the phenomenon according to which an opposing e.m.f. produce flux in a coil as a result of change in current or magnetic flux linked with a neighboring coil. But when two coils are inductively coupled, in addition to induced e.m.f. produced due to mutual induction, also induced e.m.f. is produced in each of the two coils due to self-induction.

7. (c) Presence of magnetic flux cannot produce current.

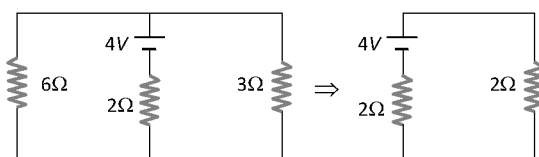
8. (d) When loop enters in field between the pole pieces, flux linked with the coil first increases (constantly) so a constant emf induces, when coil entered completely within the field, no flux change so $e = 0$. When coil exit out, flux linked with the coil decreases, hence again emf induces, but in opposite direction.

9. (b) The direction of induce current is 2 to 1 hence electron will move from 1 to 2 hence 1 is positive and 2 is negative

$$10. (c) \text{ Motional emf } e = Bvl \Rightarrow e = 2 \times 2 \times 1 = 4 \text{ V}$$

This acts as a cell of emf $E = 4 \text{ V}$ and internal resistance $r = 2\Omega$.

This simple circuit can be drawn as follows



Current through the connector $i = \frac{4}{2+2} = 1 \text{ A}$

$\therefore$ magnetic force on connector $F_m = Bil = 2 \times 1 \times 1 = 2 \text{ N}$ (Towards left)

11. In A.C. generator uses the principle of electromagnetic induction which states that when the coil is rotated in a uniform magnetic field so that the flux through it continuously changes, an emf is induced in it.

12. As current in A is increasing in clockwise direction then current induced in B is in anticlockwise direction. It seems to be the current in each section of both conductors is anti-parallel; hence they will repel each other.

13. If a loop of conducting material is moved with a constant velocity well inside a uniform magnetic field perpendicular to the field, there will not be induced current in it because to induce a current, emf induced depends upon the rate of change of magnetic flux linked with it. In this case, flux through the loop remains constant. Hence there will be no induced current in the loop.

14. Given that, initial current, $I_1 = 5.0 \text{ A}$

Final current, $I_2 = 0.0 \text{ A}$

Change in current, $dI = I_1 - I_2 = 5 \text{ A}$

Time taken for the change, $t = 0.1 \text{ s}$

Average emf, $e = 200 \text{ V}$

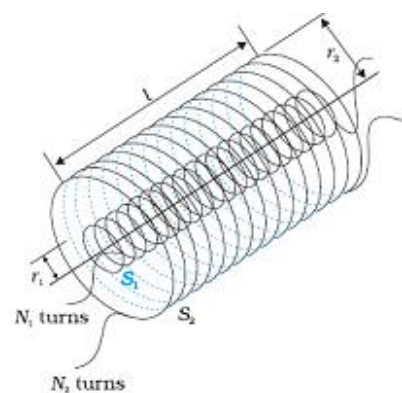
For self-inductance (L) of the coil, we have the relation for average emf as,

$$e = L \frac{dI}{dt}$$

$$L = \frac{e}{dI/dt} = 200 / \frac{5}{0.1}$$

Therefore, the self-induction of the coil is 4 H

15. Let us consider two long co-axial solenoids each of length l , as shown in figure. Let the radius of the inner solenoid S_1 be r_1 and the number of turns per unit length by n_1 . Let the corresponding quantities for the outer solenoid S_2 be r_2 and n_2 , respectively.



Let N_1 and N_2 be the total number of turns of coils S_1 and S_2 , respectively. When a current I_2 is set up through S_2 , it in turn sets up a magnetic flux through S_1 . Let us denote it by Φ_1 .

The corresponding flux linkage with solenoid S_1 is given by,

$$N_1 \Phi_1 = M_{12} I_2 \dots\dots\dots(1)$$

M_{12} is called the mutual inductance of solenoid S_1 with respect to solenoid S_2 .

It is also referred to as the coefficient of mutual induction.

The magnetic field due to the current I_2 in S_2 is $\mu_0 n_2 I_2$. The resulting flux linkage with coil S_1 is,

$$N_1 \Phi_1 = (n_1 l) (\pi r_1^2) (\mu_0 n_2 I_2) = [\mu_0 n_1 n_2 \pi r_1^2 l] I_2 \dots\dots\dots(2)$$

where $(n_1 l)$ is the total number of turns in solenoid S_1 . Thus, from Eq. (1) and Eq. (2), we get $M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l$

We now consider the reverse case. A current I_1 is passed through the solenoid S_1 and the flux linkage with coil S_2 is,

$$N_2 \Phi_2 = M_{21} I_1$$

M_{21} is called the mutual inductance of solenoid S_2 with respect to solenoid S_1 .

As done in previous case, we can also get the relation for M_{21} as

$$M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l$$

Hence if the two solenoid coils are of equal length, then $M_{12} = M_{21} = M = \mu_0 n_1 n_2 \pi r_1^2 l$

SECTION C

1-MARK QUESTIONS

Q1. In electromagnetic induction, the induced charge in a coil is independent of

- (a) Change in the flux (b) Time (c) Resistance in the circuit (d) None of the above

Q2. A coil of $40\ \Omega$ resistance has 100 turns and radius 6 mm is connected to ammeter of resistance of 160 ohms. Coil is placed perpendicular to the magnetic field. When coil is taken out of the field, $32\ \mu\text{C}$ charge flows through it. The intensity of magnetic field will be

- (a) 6.55 T (b) 5.66 T (c) 0.655 T (d) 0.566 T

Q3. A coil having n turns and resistance $R\ \Omega$ is connected with a galvanometer of resistance $4R\ \Omega$. This combination is moved in time t seconds from a magnetic field W_1 weber to W_2 weber. The induced current in the circuit is

- (a) $-\frac{W_2 - W_1}{5 Rnt}$ (b) $-\frac{n(W_2 - W_1)}{5 R t}$ (c) $-\frac{(W_2 - W_1)}{Rnt}$ (d) $-\frac{n(W_2 - W_1)}{R t}$

Q4. An infinitely long cylinder is kept parallel to an uniform magnetic field B directed along positive z axis. The direction of induced current as seen from the z axis will be

- (a) Clockwise of the +ve z axis
(b) Anticlockwise of the +ve z axis
(c) Zero
(d) Along the magnetic field

Q5. A coil has an area of $0.05\ \text{m}^2$ and it has 800 turns. It is placed perpendicularly in a magnetic field of strength $4 \times 10^{-5}\ \text{Wb/m}^2$, it is rotated through 90° in 0.1 sec. The average e.m.f. induced in the coil is

- (a) 0.056 V (b) 0.046 V (c) 0.026 V (d) 0.016 V

ASSERTION AND REASON

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.

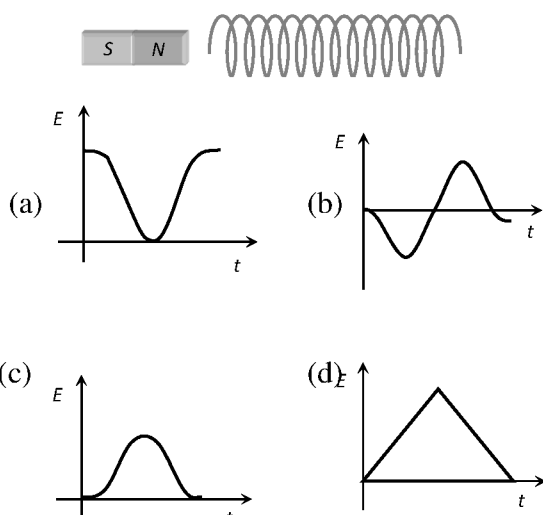
Q6. Assertion: The quantity L/R possesses dimensions of time.

Reason : To reduce the rate of increases of current through a solenoid should increase the time constant (L/R).

Q7. Assertion : When two coils are wound on each other, the mutual induction between the coils is maximum.

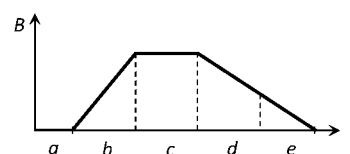
Reason: Mutual induction does not depend on the orientation of the coils.

Q8. The variation of induced emf (E) with time (t) in a coil if a short bar magnet is moved along its axis with a constant velocity is best represented as

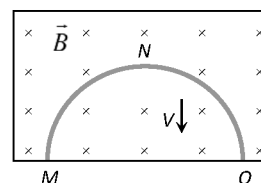


Q9. The graph gives the magnitude $B(t)$ of a uniform magnetic field that exists throughout a conducting loop, perpendicular to the plane of the loop. Rank the five regions of the graph according to the magnitude of the emf induced in the loop, greatest first

- (a) $b > (d = e) < (a = c)$ (b) $b > (d = e) > (a = c)$
 (c) $b < d < e < c < a$ (d) $b > (a = c) > (d = e)$



Q10. A thin semicircular conducting ring of radius R is falling with its plane vertical in a horizontal magnetic induction B . At the position MNQ , the speed of the ring is V and the potential difference developed across the ring is



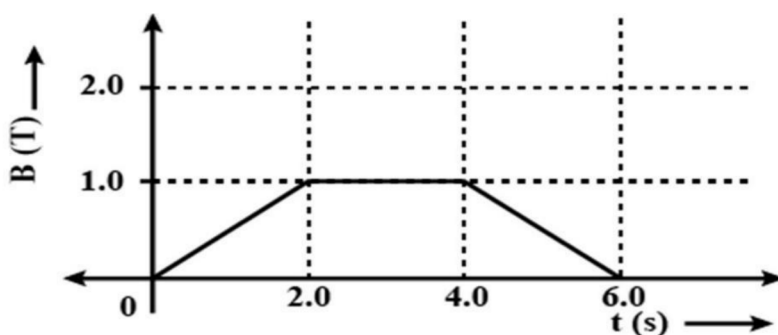
- (a) Zero (b) $Bv\pi R^2/2$ and M is at higher potential
 (c) πRBV and Q is at higher potential (d) $2RBV$ and Q is at higher potential

Q11. A conducting rod of length $2l$ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The e.m.f. induced between two ends of the rod is

- (a) $B\omega l^2$ (b) $\frac{1}{2}B\omega l^2$ (c) $\frac{1}{8}B\omega l^2$ (d) none of these

2-MARKS QUESTIONS

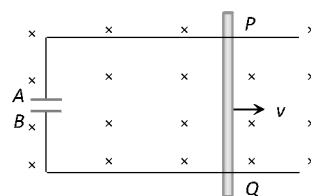
12. The magnetic field through a circular loop of wire, 12cm in radius and 8.5Ω resistance, changes with time as shown in the figure. The magnetic field is perpendicular to the plane of the loop. Calculate the current induced in the loop .



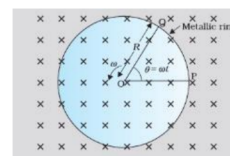
13. A short solenoid of radius a , number of turns per unit length n , and length L is kept coaxially inside a very long solenoid of radius b , number of turns per unit length n . What is the mutual inductance of the system?

3-Marks Questions

Q14. A conducting rod PQ of length $L = 1.0$ m is moving with a uniform speed $v = 2$ m/s in a uniform magnetic field $B = 4.0$ T directed into the paper. A capacitor of capacity $C = 10$ μF is connected as shown in figure. Find the charge store in capacitor.

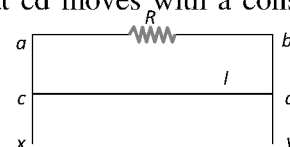


Q15. A conducting rod rotates with angular speed ω with one end at the centre and other end at circumference of a circular metallic ring of radius R , about an axis passing through the centre of the coil perpendicular to the plane of the coil. A constant magnetic field B parallel to the axis is present everywhere shown in fig. Show that the emf. between the centre and the metallic ring is $\frac{1}{2} B\omega R^2$

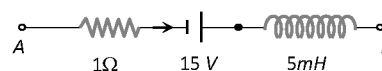


Q16. What is meant by Self Induction. Derive an expression for self-inductance which shows that it is not dependent upon emf and current induced in it.

Q17. A wire cd of length l and mass m is sliding without friction on conducting rails ax and by as shown. The vertical rails are connected to each other with a resistance R between a and b . A uniform magnetic field B is applied perpendicular to the plane $abcd$ such that cd moves with a constant velocity v . Find the required velocity.



Q18. The network shown in the figure is a part of a complete circuit. If at a certain instant the current i is 5 A and is decreasing at the rate of 10^3 A/s then find $V_A - V_B$ is



SECTION C (ANS KEY)

1-Marks Questions

1.(b) We know that $e = \frac{d\phi}{dt}$

But $e = iR$ and $i = \frac{dq}{dt} \Rightarrow \frac{dq}{dt} R = \frac{d\phi}{dt} \Rightarrow dq = \frac{d\phi}{R}$

2. (d) $q = -\frac{N}{R}(B_2 - B_1)A \cos \theta$

$$32 \times 10^{-6} = -\frac{100}{(160 + 40)}(0 - B) \times \pi \times (6 \times 10^{-3})^2 \times \cos 0^\circ$$

$$\Rightarrow B = 0.565 \text{ T}$$

3. (b) $i = \frac{e}{R} = \frac{-N}{R} \frac{(\phi_2 - \phi_1)}{\Delta t} = \frac{-n(W_2 - W_1)}{5Rt}$

4. (c) Since the magnetic field is uniform therefore there will be no change in flux hence no current will be induced.

5. (d) $e = -\frac{NBA(\cos \theta_2 - \cos \theta_1)}{\Delta t}$

$$= -\frac{800 \times 4 \times 10^{-5} \times 0.05 (\cos 90^\circ - \cos 0^\circ)}{0.1} = 0.016 \text{ V}$$

6. (b) The relation of induced emf is $e = \frac{L di}{dt}$ and current i is given by $i = \frac{e}{R} = \frac{1}{R} \cdot \frac{L di}{dt} \Rightarrow$

$$\frac{di}{dt} = i \frac{R}{L} = \frac{i}{L/R}.$$

In order to decrease the rate of increase of current through solenoid. We have to increase the time constant $\frac{L}{R}$.

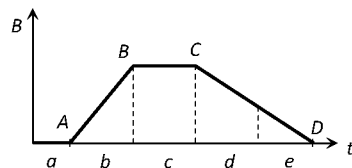
7. (c) The manner in which the two coils are oriented, determines the coefficient of coupling between them.

$$M = K^2 \cdot L_1 L_2$$

When the two coils are wound on each other, the coefficient of coupling is maximum and hence mutual inductance between the coil is maximum.

8. (b) As the magnet moves towards the coil, the magnetic flux increases (nonlinearly). Also there is a change in polarity of induced emf when the magnet passes on to the other side of the coil.

9. (b) Induced emf $e = A \frac{dB}{dt}$



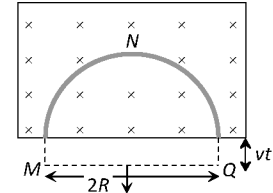
i.e. $e \propto \frac{dB}{dt}$ (= slope of B – t graph)

In the given graph slope of AB > slope of CD, slope in the ‘a’ region = slope in the ‘c’ region = 0, slope in the ‘d’ region = slope in the ‘e’ region $\neq 0$. That’s why $b > (d = e) > (a = c)$

10. (d) Rate of decrease of area of the semicircular ring $-\frac{dA}{dt} = (2R) v$

According to Faraday's law of induction induced emf

$$e = -\frac{d\phi}{dt} = -B \frac{dA}{dt} = -B (2Rv)$$



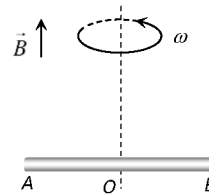
The induced current in the ring must generate magnetic field in the upward direction. Thus Q is at higher potential.

11.(d) Potential difference between

O and A is $V_0 - V_A = \frac{1}{2} B l^2 \omega$

O and B is $V_0 - V_B = \frac{1}{2} B l^2 \omega$

so $V_A - V_B = 0$



2-MARKS QUESTIONS

12. $\epsilon = -d\phi/dt$

$\epsilon = -d(BA)/dt$

$\epsilon = -AdB/dt$

$0 < t < 2$ $I = -(3.14 \times 0.12 \times 0.12 \times 1)/2 \times 8.5 = -0.0026 \text{ A}$

$2 < t < 4$ $I = 0$

$4 < t < 6$ $I = 0.0026 \text{ A}$

13. $M = \mu_0 n_1 n_2 \pi a^2 L$

3-Marks Questions

14. $Q = CV = C (Bvl) = 10 \times 10^{-6} \times 4 \times 2 \times 1 = 80 \mu\text{C}$

According to Fleming's right hand rule induced current flows from Q to P. Hence P is at higher potential and Q is at lower potential. Therefore A is positively charged and B is negatively charged.

15. As the rod is rotated, free electrons in the rod move towards the outer end due to Lorentz force and get distributed over the ring. Thus, the resulting separation of charges produces an emf across

the ends of the rod. At a certain value of emf, there is no more flow of electrons and a steady state is reached. Using Eq. (6.5), the magnitude of the emf generated across a length dr of the rod as it moves at right angles to the magnetic field is given by

$$dE = BV \, dr. \text{ Hence } E = \int dE = \int BV \, dr = \int_0^R BV \, dr$$

$$V = \omega r$$

$$E = B\omega \int_0^R r \, dr = \frac{B\omega R^2}{2}$$

16. Coefficient of Self-Induction

Consider current I flowing through a long solenoid of area A ,

Let N be the total number of turns in the solenoid,

$$\text{Total flux, } \phi = NBA$$

$$\text{Here, } B = \mu_0 n I$$

Where, n is no. of turns per unit length of the solenoid

$$N = nl \Rightarrow \phi = nl \times \mu_0 n I A$$

$$\Rightarrow \phi = \mu_0 n^2 A l I \dots\dots\dots(1) \text{ Also,}$$

$$\phi = LI \dots\dots\dots(2)$$

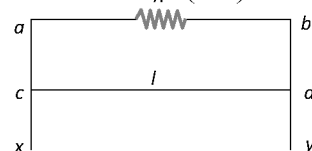
$$\text{From equation (1) \& (2) } \mu_0 n^2 A l = L \Rightarrow L = \mu_0 n^2 A l$$

$$L = \mu_0 N^2 A / l \text{ --- this equation show that it is independent on emf and current}$$

17. (b) Due to magnetic field, wire will experience an upward force $F = B i l = B \left(\frac{B v l}{R} \right) l \Rightarrow F = \frac{B^2 v l^2}{R}$

If wire slides down with constant velocity then

$$F = mg \Rightarrow \frac{B^2 v l^2}{R} = mg \Rightarrow v = \frac{mgR}{B^2 l^2}$$



18. (c) By using Kirchoff's voltage law

$$V_A - iR + E - L \frac{di}{dt} = V_B \Rightarrow V_B - V_A = 15 \text{ volt.}$$

